

Does the MOND Acceleration Scale Evolve? A Pre-Registered Decisive Measurement: the Deep-MOND Tully–Fisher Zero Point of One Lensed Rotator at $z \simeq 2.5$, with a Two-Stage JWST/ALMA Funnel

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Abstract

The de Sitter–MOND framework ties the galactic acceleration scale to the dark-energy density, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, and therefore predicts that a_0 does not evolve while Λ does not: the deep-MOND baryonic Tully–Fisher zero point is the same at $z = 2.5$ as at $z = 0$ to better than 1%. The Λ CDM-native alternative, an emergent acceleration scale set by halo structure, rises with redshift, $a_s(z)/a_s(0) = E(z)^{4/3} [c^2/f(c)](z)/[c^2/f(c)](0)$, a factor 2.1 at $z = 2.5$, i.e. a zero-point displacement of 0.33 dex in baryonic mass at fixed velocity or 20% in velocity at fixed mass. We show, from a twelve-constraint joint likelihood over every published high-redshift Tully–Fisher zero point and the 21-object ledger of candidate deep-MOND targets, that the existing archives cannot decide this: every published sample above $z = 0.5$ except one lensed-dwarf sample sits at $1.7\text{--}6.4 a_0$, where the zero point keeps only 7–18% of the a_0 lever and is degenerate with disc-size evolution, and no known object passes the deep-MOND, rotation and lens-quality gates simultaneously. We therefore pre-register the measurement that would: one strongly lensed, rotationally supported galaxy at $2.3 \lesssim z \lesssim 2.9$ with $g_{\text{bar}}(R_{\text{out}}) < 0.3 a_0$ on both a_0 footings and at the upper end of its baryonic-mass uncertainty, screened for rotation with the JWST NIRSpec IFU (G235H/F170LP places $\text{H}\alpha$ and $[\text{O III}]$ in one setting for $2.32 < z < 3.83$) before ALMA time is spent, and then given CO(3–2) in ALMA Band 3 ($1.98 < z < 3.12$) for an independent gas mass and a second dynamical tracer. The decision statistic $\Delta_{\text{BTFR}} = \log_{10} M_b - 4 \log_{10} V_f - C_0$ with C_0 frozen from the local calibration is scored against the two pre-specified values, 0.00 and +0.33 dex, at a required total uncertainty of 0.13 dex, which gives 20:1 discrimination. No drift nuisance may be introduced after the target is observed; a result inconsistent with both values counts against both. Every number is produced by a committed script.

1 The two predictions

The framework’s law $a_0(z)/a_0(0) = \sqrt{\rho_{\text{DE}}(z)/\rho_{\text{DE}}(0)}$ is flat for a cosmological constant and within 1% of flat for $z \leq 5$; on the DESI DR2 $w_0 w_a$ fit it declines to 0.82 at $z = 2.5$ (−0.09 dex). In deep MOND $V_f^4 = G a_0 M_b$ exactly, so the baryonic Tully–Fisher zero point at fixed velocity moves by $\Delta_{\text{BTFR}} = -\log_{10}[a_0(z)/a_0(0)]$. Λ CDM has no fundamental a_0 ; its radial-acceleration scale is emergent from halo structure and rises, for the Dutton–Macciò 2014 concentration–mass relation at $M = 10^{12} M_\odot$, by factors 1.23, 1.76, 2.13 and 2.82 at $z = 1, 2, 2.5$ and 3.25 (Duffy 2008: 1.43, 2.28, 2.80, 3.67; the Magneticum hydrodynamical apparent rise $(1+z)^{0.92}$: 1.89, 2.75, 3.17, 3.79). At

$z \simeq 2.5$ the two predictions are

$$\Delta_{\text{BTFR}}^{\text{framework}} = 0.00 \text{ (} -0.09 \text{ with DESI dark energy)}, \quad \Delta_{\text{BTFR}}^{\Lambda\text{CDM}} \simeq +0.33 \text{ dex}, \quad (1)$$

equivalently $V_{\Lambda\text{CDM}}/V_{\text{framework}} = 2.1^{1/4} = 1.20$ at fixed baryonic mass: for a dwarf at 60 km s^{-1} the competing prediction is 72 km s^{-1} . A single point with total zero-point uncertainty 0.13 dex separates them at 20:1. Both a_0 footings (9.3619×10^{-11} and $1.1279 \times 10^{-10} \text{ m s}^{-2}$) are carried; the ratio $a_0(z)/a_0(0)$ is footing-independent.

2 Why the archives cannot decide

The repository’s joint likelihood over twelve published constraints (Übler et al. 2017 at $z \simeq 0.9$ and 2.3; Tiley et al. 2016, 2019; Pelliccia et al. 2017; Jeanneau et al. 2026 lensed dwarfs at $z \simeq 0.9$; Amvrosiadis et al. 2025; Straatman et al. 2017; Turner et al. 2017; Danhaive et al. 2026; the Big Wheel at $z = 3.25$; MSA-3D; Ciocan et al. MUSE-DARK III; KLASS) returns an undecided, prior-dominated verdict: the Bayes factor between the flat law and the ΛCDM -native rise flips sign with the ceiling placed on an apparent-drift nuisance (beam smearing, pressure support, selection, unmeasured gas), and the worst-case $|\log_{10} B|$ over that ladder is 0.19 where 20:1 needs 1.30. The reason is physical rather than statistical. Every published sample above $z = 0.5$ except the Jeanneau lensed dwarfs ($g_{\text{bar}} \approx 0.3 a_0$) and the Big Wheel sits at $g_{\text{bar}} = 1.7\text{--}6.4 a_0$, where the deep-MOND map does not apply, only 7–18% of the a_0 lever survives, and the zero point is degenerate with the observed disc-compactness evolution $R \propto (1+z)^{-0.75}$: the concordant -0.27 dex of Übler and Amvrosiadis at $z \simeq 2.3$ is simultaneously rising- a_0 -shaped, ΛCDM -halo-shaped and constant- a_0 -plus-size-shaped. The committed ledger of 21 candidate deep-MOND objects at $1.5 < z < 3.3$, scored on both footings by deep-MOND probability, rotational support, lens-model quality, gas feasibility and source-plane resolution, contains no object that passes the flagship gate: every measured rotator is high-acceleration and every deep-MOND candidate has no measured rotation. The most magnified gas-measured object, the MACS J0451+0006 arc at $z = 2.01$, is dispersion-dominated ($V \sin i/\sigma = 0.48$) with two refereed lens models disagreeing by a factor 2.3, and is retained only as a gas-analysis control. The decisive object has to be found and measured.

3 The target gate

A galaxy enters the decisive sample only if source-plane modelling establishes all of the following:

1. $g_{\text{bar}}(R_{\text{out}}) < 0.30 a_0$ on both footings and at the upper end of the baryonic-mass uncertainty, with g_{bar} reconstructed from the resolved stellar and gas distributions, not a point mass; preferred $< 0.20 a_0$;
2. $V_{\text{rot}}/\sigma > 1.5$ from a forward-modelled source-plane disc fit with intrinsic dispersion, inclination, beam smearing, asymmetric drift, clumps and outflow components fitted simultaneously; preferred > 2 ; a regular monotonic velocity field;
3. $\delta \log_{10} \mu < 0.08 \text{ dex}$ and no credible alternative lens reconstruction differing by more than the adopted systematic budget;
4. inclination $35^\circ < i < 75^\circ$ and at least five independent source-plane resolution elements along the kinematic major axis;

5. redshift $2.3 \lesssim z \lesssim 2.9$, so that $H\alpha$ and $[\text{O III}]$ fall together in NIRSpec G235H/F170LP (1.66–3.17 μm , $R \sim 2700$) and CO(3–2) at 345.8 GHz rest falls in ALMA Band 3 (84–116 GHz; $[\text{C II}]$ at $z \simeq 2.5$ lands near 543 GHz, in the gap between Bands 8 and 9).

A galaxy failing the rotation or deep-MOND criterion is rejected without interpretation as a Tully–Fisher test. The candidate pool is the faint, highly magnified population that currently lacks resolved kinematics (public examples include Abell 68 C4 at $z = 2.622$ and C20b at $z = 2.689$, neither of which is claimed to pass); the strategy is to start from a genuinely low-mass lensed galaxy and prove it rotates, not to take a known rotator and hope it is deep-MOND.

4 The two-stage funnel

Stage 1, NIRSpec IFU. Observe the candidates deeply enough to recover a two-dimensional velocity field and construct a forward-modelled source-plane velocity cube rather than de-lensing a derived rotation curve. Advance a candidate only if it passes the rotation gate and if $g_{\text{bar},*}(R_{\text{out}}) + g_{\text{gas,max}}(R_{\text{out}}) < 0.3 a_0$ using a conservative upper estimate of the still-unmeasured gas, so that ALMA time is never spent on a galaxy that could not discriminate a_0 .

Stage 2, ALMA CO(3–2). Recover the integrated molecular-gas mass and a resolved CO velocity field. The decisive galaxy must have two independent dynamical tracers; a material CO/optical dynamical discrepancy invalidates the target rather than being absorbed into a larger uncertainty. The baryonic model is $M_b = M_* + 1.36(M_{\text{HI}} + M_{\text{H}_2})$ with the CO excitation and X_{CO} uncertainties propagated explicitly.

5 The pre-registered statistic and its budget

The estimator is the one frozen in the repository’s target list: galaxy median then median across galaxies (a generalised-least-squares combination is forbidden; it was shown to carry a +10 percentage-point bias). The statistic is $\Delta_{\text{BTFR}} = \log_{10} M_b - 4 \log_{10} V_f - C_0$ with C_0 frozen from the local calibration before the high-redshift result is examined. The program succeeds only if $\sigma(\Delta_{\text{BTFR}}) \leq 0.13$ dex; because $M_b \propto V^4/a_0$, that is 7.8% in velocity if velocity were the only uncertainty, so the proposal requirements are $\delta V_{\text{rot}}/V < 5\%$ from the combined tracers and $\delta \log_{10} M_b < 0.10$ dex, leaving 0.05–0.07 dex for inclination, lens reconstruction and asymmetric drift. The lensing requirement, $\delta \log_{10} \mu \lesssim 0.05\text{--}0.08$ dex, is therefore severe and by itself removes MACS J0451 from flagship status. Controls: MACS J0451 as a gas-analysis control; the Cosmic Eye at $z = 3.074$, with resolved rotation and published molecular gas, as an end-to-end pipeline validation object (it is not deep-MOND once its stellar and gas mass are included).

6 Decision rule

A result near 0.00 dex supports the framework’s frozen constancy; a result near +0.33 dex supports the Λ CDM-native rise; a result inconsistent with both at the stated uncertainty is evidence against both and is reported as such. No drift nuisance is introduced after the target is observed. Any other significant displacement falsifies the framework’s low-redshift constancy prediction. One galaxy is enough if and only if it is genuinely deep-MOND, rotation-dominated, independently gas-measured and lens-model-controlled.

Reproducibility and provenance

The joint likelihood, the Λ CDM-native law and the ledger are in `prep_2026/a0z_crossscale/` (`a0z_fork_likelihood_2026.py`, `a0z_lcdm_native_hypothesis_2026.py`, `highz_deepmond_target_list_2026.py`, `highz_target_score_2026.py`) and `prep_2026/highz_tfr_fork/` of the public repository; the a_0 footings and the flat law are those of the framework’s standing record. The funnel ordering, the CO(3–2)/Band-3 choice and the precision budget were proposed in an exchange with an OpenAI model on 6 September 2026 and are adopted here after checking against the ledger; the ledger’s verdict that no existing object passes the gate is unchanged by the score. The framework’s coefficient $\kappa = \frac{1}{2}$ is fitted, not derived (DOI 10.5281/zenodo.22559892); this measurement tests the law’s redshift dependence, which is coefficient-blind.